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Part 1:

Original Prompt V1:

Attached product-reviews.csv

“Are these reviews positive or negative?”

Response:

“The reviews are split almost evenly. Not many of them are clearly one or the other:

Sentiment	***Count***
Negative	*37*
Positive	*36*
Mixed	*20*
Neutral	*17*

By product:

Product	***Pos***	***Neg***	***Mixed***	***Neutral***
Lumen Smart Bulbs	*7*	*9*	*3*	*3*
Hearth Kettle	*7*	*8*	*4*	*3*
Trailmark Backpack	*7*	*8*	*4*	*3*
Breeze Earbuds	*7*	*7*	*4*	*4*
Cozy Knit Blanket	*8*	*5*	*5*	*4*

The blanket does the best. The smart bulbs do the worst, mostly because of complaints about the app and Wi-Fi connection.

How I labeled the tricky ones:

- ***Sarcasm:*** *I counted 10 sarcastic reviews as negative, like “Love that the waterproof pack turned my sleeping bag into a sponge. Ten stars.” A keyword-based sentiment tool would probably score those as positive. They’re flagged in the note column.*

- ***Mixed:*** *these reviews have real praise and a real complaint, like "Great bulbs, terrible app." If you need a strict positive/negative split, most of them lean one way and I can force them.*
- ***Neutral:*** *this covers questions ("Does this work with PS5?"), reviews that are only about shipping or the seller ("Shipping took 3 weeks"), and flat ones ("It's a kettle. It boils water.").*

The full labeled list is in product-reviews-labeled.csv."

*Also attached a .csv file called 'Product reviews labeled' with its labeling of each review and its reasoning why it did so.

The LLM had a response that went ahead and made its own .csv file, labeling each review with accurate labels as well, catching onto sarcasm and such. The results were presented with tables showcasing the products with their pos, neg, mixed, and neutral. It also gave a total of each sentiment. The judgement it had about each review was accurate. It labeled every review individually instead of just summarizing, and because it put the labels in a CSV, I could sort and count the results. It handled the unclear reviews by adding Mixed and Neutral categories on its own, which I never asked for.

Part 2:

Improved Prompt V2 | Output Format & Role Prompting:

attached file

" You are someone working for a review company and are building reports for other companies who have a bunch of reviews in. You are deciding on a companies general troubles along with their strong suits.

I uploaded a file for you to look at. You are going to look through it and sort the reviews based on their reasoning for the review.

Find the common theme between reviews and figure out what is driving people to buy or not buy the items.

The output should only be the companies troubles and strong suits with the product along with its reasoning. Be concise and only write / deliver whats needed. Also, please give % of each style review."

Response:

"Review breakdown (110 reviews)

Type	*Share*
Positive	*31.8%*
Negative	*23.6%*
Sarcastic (negative)	*9.1%*
Mixed	*19.1%*
Neutral / no opinion	*7.3%*
Not about the product (questions, shipping, seller errors)	*9.1%*

About a third of reviews are negative once the sarcastic ones are counted, roughly the same as the positive share.

By product

Lumen Smart Bulbs: 32% positive, 41% negative. This is the weakest product.

- *Problems: The software lets down the hardware. People report the app crashing, Wi-Fi dropouts, settings wiped by updates, a required account and location data, buzzing when dimmed, dead bulbs, and brightness that falls short of what's claimed.*
- *Strengths: The features people like are sunrise alarm, scheduling, music sync and Alexa voice control. Owners who get through setup buy more, and support fixes pairing problems quickly.*
- *What drives the decision: People love the bulbs and hate the app.*

Hearth Electric Kettle: 32% positive, 36% negative.

- *Problems: There are safety and build issues: it leaks from the base, the handle gets very hot, the lid won't stay shut, and the switch fails early. Smaller complaints are a plastic taste, mineral buildup, a short cord, a 30-minute keep-warm, a loud beep and small capacity.*

- *Strengths: It boils fast and looks good on the counter. Tea drinkers like the temperature presets and coffee drinkers like the gooseneck spout. Customer service is fast and sends replacements.*
- *What drives the decision: People buy it for the look and speed. Defects and safety are what lose them, and customer service is what wins them back.*

Breeze Wireless Earbuds: 27% positive, 32% negative.

- *Problems: Hardware reliability is the main issue: charging fails, there's static, connections drop, and the case lid snaps. The mic sounds muffled, the touch controls get triggered by accident, and fit is inconsistent. The listing implies noise cancelling the earbuds don't have, and support doesn't respond.*
- *Strengths: Value is the big one, with frequent comparisons to AirPods and $200 earbuds. People also like the battery life, comfort, instant pairing and sound quality for the price.*
- *What drives the decision: People buy them because they're cheap and sound good. They return them because the hardware fails.*

Trailmark Hiking Backpack: 32% positive, 32% negative.

- *Problems: The listing doesn't match the product: it's not waterproof, it's too small for anything past day trips, and the color differs from the photos. Quality is inconsistent (zipper, buckle and stitching failures). Other complaints are a chemical smell, sore shoulders from the straps and a thin hip belt.*
- *Strengths: It's durable and feels premium for the price. It works well for day hikes and school, and people like the pockets and hydration sleeve.*
- *What drives the decision: Many units are excellent, but a run of defective ones and an inaccurate listing are pulling the rating down.*

Cozy Knit Throw Blanket: 36% positive, 23% negative. This is the strongest product.

- *Problems: Material quality varies: shedding, itching, snagging, unraveling, shrinking in the wash and a warehouse smell. It's thinner and smaller than people expect, and the listing leaves out the dimensions.*
- *Strengths: It's soft and looks expensive. It holds up to washing and kids, and people buy repeats and gifts.*
- *What drives the decision: It's a good-looking gift at a good price. It loses buyers when it's thinner or smaller than they expected.*

Problems across all five products

1. *Listings that oversell. Waterproofing, noise cancelling, brightness, size and color are all claimed in ways reviewers say aren't true, and many complaints come down to expectations not being met.*
2. *Early failures. A lot of reviews describe products breaking within days or weeks. This is the biggest driver of returns.*
3. *Fulfillment. About 9% of reviews are about wrong items, damaged boxes or slow shipping rather than the product, and these drag down the ratings.*
4. *Value is the main reason people buy. "Feels more expensive than it was" comes up for every product.*

Reviews that praised and criticized the product equally, or that sounded positive but were sarcastic, are counted as Mixed or Sarcastic. Sarcastic reviews are counted as negative in the per-product numbers."

Improved Prompt V3 | Define the categories & Output Format:

attached file
"You are going to go through this file and label all reviews into 2 categories. Positive and negative. No neutral. No other. No exceptions.

Positive is a good word in for the product. Negative is a bad word in for the product.

Your answer should be very short and to the point with the information I asked for, the P and N sorting.

Do not veer and continue with your own instructions or way or doing things. Follow my guideline."

Response:

"P (58): R002, R006, R007, R009, R010, R012, R014, R016, R017, R020, R021, R023, R026, R027, R029, R033, R034, R035, R036, R037, R039, R040, R041, R042, R043, R044, R045, R046, R047, R050, R052, R054, R055, R056, R059, R061, R064, R067, R069, R070, R072, R077, R079, R081, R082, R083, R084, R086, R088, R089, R090, R092, R094, R101, R102, R104, R105, R106

N (52): R001, R003, R004, R005, R008, R011, R013, R015, R018, R019, R022, R024, R025, R028, R030, R031, R032, R038, R048, R049, R051, R053, R057, R058, R060, R062, R063, R065, R066, R068, R071, R073, R074, R075, R076, R078, R080, R085, R087, R091, R093, R095, R096, R097, R098, R099, R100, R103, R107, R108, R109, R110

R012, R067, R094 and R101 say nothing good or bad, so I had to pick for them."

Part 3:

Re-run V1: (My guess on left (P – Positive | N – Negative | M - Mixed)

"Are these reviews positive or negative?

M *R043,Hearth Electric Kettle,"It's fine. Does the job, nothing special."*

P *R045,Cozy Knit Throw Blanket,Perfect weight for summer evenings on the porch.*

P *R089,Trailmark Hiking Backpack,"Received it today, looks nice."*

M *R093,Cozy Knit Throw Blanket,Delivery driver left it in the rain.*

M *R026,Lumen Smart Bulbs (4-Pack),Package arrived a day early.*

M *R014,Hearth Electric Kettle,Mine arrived dented and customer service shipped a new one the next day. The kettle itself is fine.*

P *R009,Lumen Smart Bulbs (4-Pack),"Great bulbs, terrible app. If you can get past setup you'll love them."*

P *R033,Breeze Wireless Earbuds,Best purchase I've made all year. My old $200 pair is collecting dust now.*

P *R023,Lumen Smart Bulbs (4-Pack),Sunrise alarm feature has made my mornings so much easier.*

N *R022,Trailmark Hiking Backpack,"Love that the ""waterproof"" pack turned my sleeping bag into a sponge. Ten stars."*

Response:

"Mostly positive, but a few depend on whether you count delivery and customer service or only the product:

ID	*Product*	*Verdict*	*Why*
R043	*Hearth Electric Kettle*	*Neutral*	*"Fine, nothing special." It works, but there's no real praise.*

ID	*Product*	*Verdict*	*Why*
R045	*Cozy Knit Throw Blanket*	*Positive*	*Straightforward praise*
R089	*Trailmark Hiking Backpack*	*Neutral*	*They got it today and haven't used it yet. "Looks nice" is only a first impression.*
R093	*Cozy Knit Throw Blanket*	*Negative (shipping)*	*The complaint is about the driver, not the blanket*
R026	*Lumen Smart Bulbs*	*Positive (shipping)*	*Says nothing about the bulbs themselves*
R014	*Hearth Electric Kettle*	*Mixed*	*Arrived damaged, but service fixed it fast. The product itself is just "fine."*
R009	*Lumen Smart Bulbs*	*Mixed*	*They love the bulbs and hate the app*
R033	*Breeze Wireless Earbuds*	*Positive*	*Strong praise*
R023	*Lumen Smart Bulbs*	*Positive*	*Names a specific feature that helps them*
R022	*Trailmark Hiking Backpack*	*Negative*	*Sarcasm. The "waterproof" pack soaked their sleeping bag, and "Ten stars" is a joke.*

Tally: 4 positive, 2 negative, 2 neutral, 2 mixed.

R022 is the one a basic sentiment tool would get wrong, since "Love" and "Ten stars" make it look like a rave. If you're scoring product quality, set R093 and R026 aside because they're only about shipping."

Spot-check table (my labels vs. Version 3, my best prompt)

Review	My label	Model label (V3)	Agree?
R043	Mixed/Neutral	Positive	No
R045	Positive	Positive	Yes
R089	Positive	Positive	Yes
R093	Mixed/Neutral	Negative	No

Review	My label	Model label (V3)	Agree?
R026	Mixed/Neutral	Positive	No
R014	Mixed/Neutral	Positive	No
R009	Positive	Positive	Yes
R033	Positive	Positive	Yes
R023	Positive	Positive	Yes
R022	Negative	Negative	Yes

We agreed on 6 of 10. Every disagreement is a review I labeled mixed or neutral that V3 had to force into positive or negative, because that prompt only allowed two labels.

I also re-ran the original prompt on just these 10 reviews in a fresh chat to see how Claude labeled them when it was allowed to use mixed and neutral. That run gave 4 positive, 2 negative, 2 neutral and 2 mixed, much closer to my own labels.

Who was right on the disagreements:

- **R043, R014:** I think I'm right. "It's fine" and "the kettle itself is fine" aren't really positive. V3 only labeled them positive because it wasn't allowed to say neutral or mixed.
- **R093, R026:** These are only about shipping, not the product. Neither positive nor negative is right for them. In the re-run, Claude also pointed out they should be set aside when scoring the product.
- **R009 ("Great bulbs, terrible app"):** I labeled this positive, but looking again, the model's re-run label of Mixed is more accurate. It has clear praise and a clear complaint.
- **R089 ("Received it today, looks nice"):** I'm keeping mine as positive. The re-run called it neutral because they hadn't used it yet, but they still said something good about it.

What would fix it: add a third category to the prompt with a clear definition, e.g. "Neutral: the review is only about shipping or the seller, or gives no real opinion of the product." Adding one few-shot example, like "Delivery driver left it in the rain." → Neutral, would show Claude exactly how to handle those.

It shocked me seeing AI decided a review that mentioned a bad delivery driver, led to a negative review as it had nothing to do with the product itself. Also, the review saying "it looks nice", in my eyes is a positive review as they like the product. Overall, I am a bit upset

with Ai's judgement here reviewing the examples. I think it needs a bit more machine learning sessions to interpret human feelings.

Analysis:

I am very surprised with these test cases. When using a CSV data file including reviews, there is a ton of ways to work on and structure the output so I figured I would give it two wildly different scenarios.

The first scenario was written to see what the LLM would do if given instructions to act as someone working for a review company, building reports for those who have reviews. I did not want to just simply sort the reviews into 3 groups.
Instead, I wanted to see if the reviews on items had relations between one another with the reason why they are buying or not buying the product. Claude did quite a good job looking through and picking out information regarding each item and placing Them within one of three sections: Problems, Strengths, What drives the decision.

Overall, I was happy with the results of the V2 prompt.

Heading over into V3, we switched into a different mode to get some results based on what is clearly guessing. I had forced the AI to choose between either positive or negative review. No in between. No neutral. No exceptions.

The results were interesting. It placed 58 into positive and 52 in negative. It did not fold under pressure when deciding for countless reviews and it said they had to pick for them. It ruined the count without thinking twice. Interesting.

In V2, I used techniques of output format and role prompting to get the interesting and complex output of the categorized reviews along with the proper percentages for each product.

In V3, I used define the categories and output format. I had to use define the categories because in order to shape the response to be in only two categories, I had to explicitly state that there are ONLY two, P and N. I also made sure the AI gave a very short response in which it did. It gave only the shortest possible label for each review, for example 'R059'. I am surprised it went ahead and sorted the reviews with CLEARLY no positive or negative attitude… usually Claude can be stubborn and hate making mistakes, but it allowed for it.

Did it label every review? Yes. V1 and V3 both labeled all 110 reviews. V2 didn't label them one by one. It gave percentages instead.

Tricky reviews: Claude caught sarcasm every time, like R022 ("Ten stars"). It struggled with reviews that had no real opinion or were only about shipping. In V3 it had to guess on those, and that caused all 4 disagreements in my spot-check. Adding a Neutral category would fix it.

Biggest difference: Defining the categories. It changed the output the most, and it caused every mistake in my spot-check.